

CAPSTONE PROJECT / ENTERPRISE AI RAG SUPPORT

SCHOOLTRUST AI

Verified Parent Support and Knowledge Improvement System

A six-minute capstone walkthrough with one complete improvement loop.

Project	SchoolTrust AI
Student	Moeen Ahmad Butt
Recommended duration	6 to 8 minutes
Required browser tabs	Web interface, n8n workflow and Supabase tables
Version / date	Version 1.0 14 July 2026

DEMO PRESENTATION SCRIPT

One focused project. Approved sources. Human-controlled improvement.

1. Opening - 40 seconds

Say this

My project is SchoolTrust AI, a verified parent-support and knowledge-improvement system. The problem is not simply answering school questions. The real problem is preventing outdated or incomplete answers and improving the knowledge base when the system cannot answer safely.

2. Show the Real Problem - 45 seconds

- Explain that a normal chatbot may give only one annual-fund amount.
- State that the correct answer depends on whether the student is a new admission or a promoted student.
- Explain that generic fallback responses also waste the knowledge already available.

3. Verified Answer - 60 seconds

Action	What to explain
Ask: What is the annual fund?	The workflow classifies fees as critical and applies the higher confidence requirement.
Show the response	The assistant states both categories or asks which category applies.
Point to the status label	Verified means the answer cited an active approved source and passed guardrails.

4. Knowledge Failure Recovery - 90 seconds

Action	What to explain
Ask: What is the current transport fee?	No approved transport source exists.
Show safe escalation	The AI does not invent an amount.
Open Gap Review	The unsupported question is recorded with topic, reason, priority and occurrence count.
Ask again if useful	A repeated identical question increments the open gap instead of creating uncontrolled duplicates.

5. Human Approval - 75 seconds

Action	What to explain
Open Review and Approve	The administrator enters the exact authorised answer and effective dates.
Publish	n8n creates the embedding, stores the approved record, resolves the gap and writes an audit action.
Run Manual Quality Test	Critical questions are retested after the knowledge change.

6. Close the Loop - 45 seconds

- Ask the same transport-fee question again.
- Show that the system now retrieves the administrator-approved source.
- Open Knowledge Health and show test pass rate and gap count.
- State that this is the distinctive part: question, verification, safe failure, human approval and retesting.

7. Technical Summary - 45 seconds

Layer	Implementation
Interface	Responsive HTML, CSS and JavaScript parent and administrator views.
Automation	One n8n workflow with 38 nodes and four logical modules.
Data	Supabase PostgreSQL, pgvector, RLS and server-side RPC functions.
AI	Embeddings for retrieval and a controlled JSON response for answer generation.
Safety	Approved-source filtering, effective dates, source validation, critical thresholds and human approval.

8. Likely Assessor Questions

Question	Answer
Is this just a chatbot?	No. The chatbot is one interface. The core project is the answer-assurance and knowledge-improvement automation.
Does it train itself?	No. It records gaps and proposes a workflow. An authorised administrator approves every new source.
Why use AI?	AI handles language understanding, semantic retrieval and controlled response generation. Safety decisions remain rule-based.
What happens if sources conflict?	The workflow clarifies valid categories or escalates incompatible information. It does not choose one silently.
Can it access student records?	Not in this project. Public information and private student systems are intentionally separated.
What makes it student-feasible?	It is limited to one school, three webhooks, one database and a controlled set of approved documents and tests.

Final sentence

SchoolTrust AI does not only answer what the school already knows. It shows the school what its support system still needs to know, and improves only after human approval.